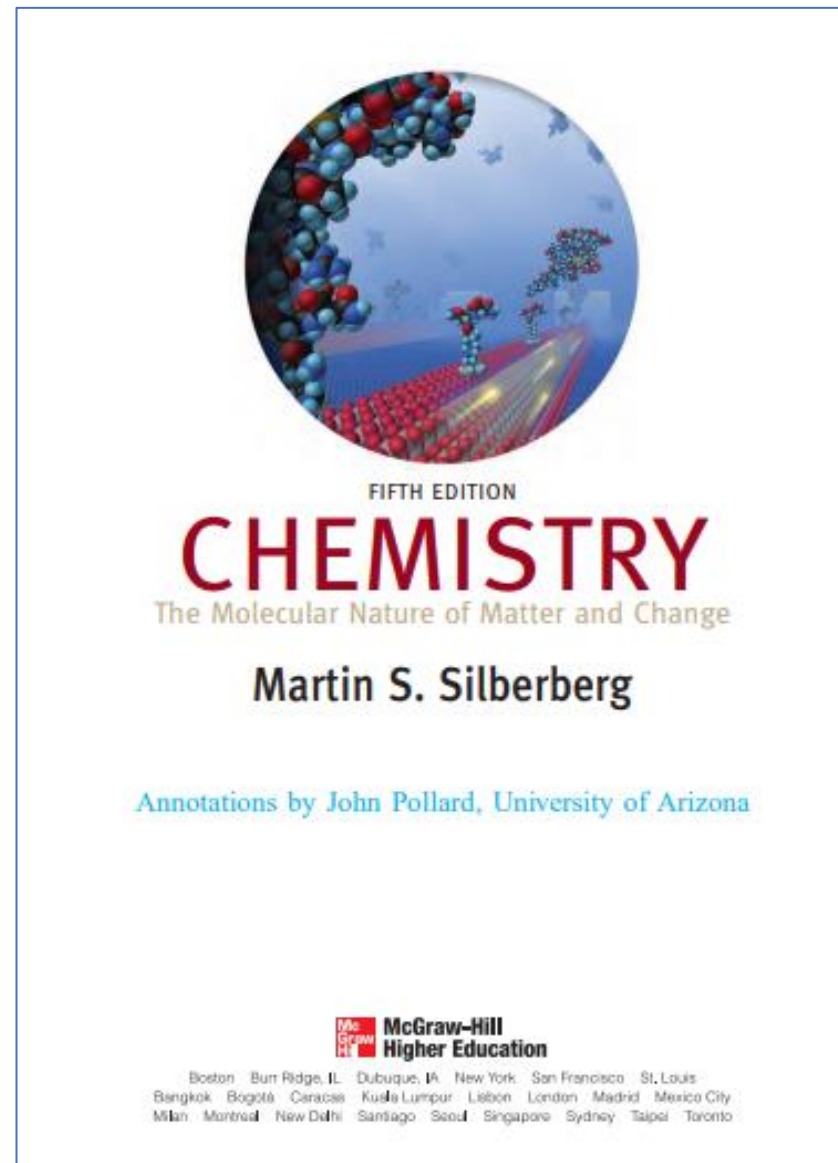
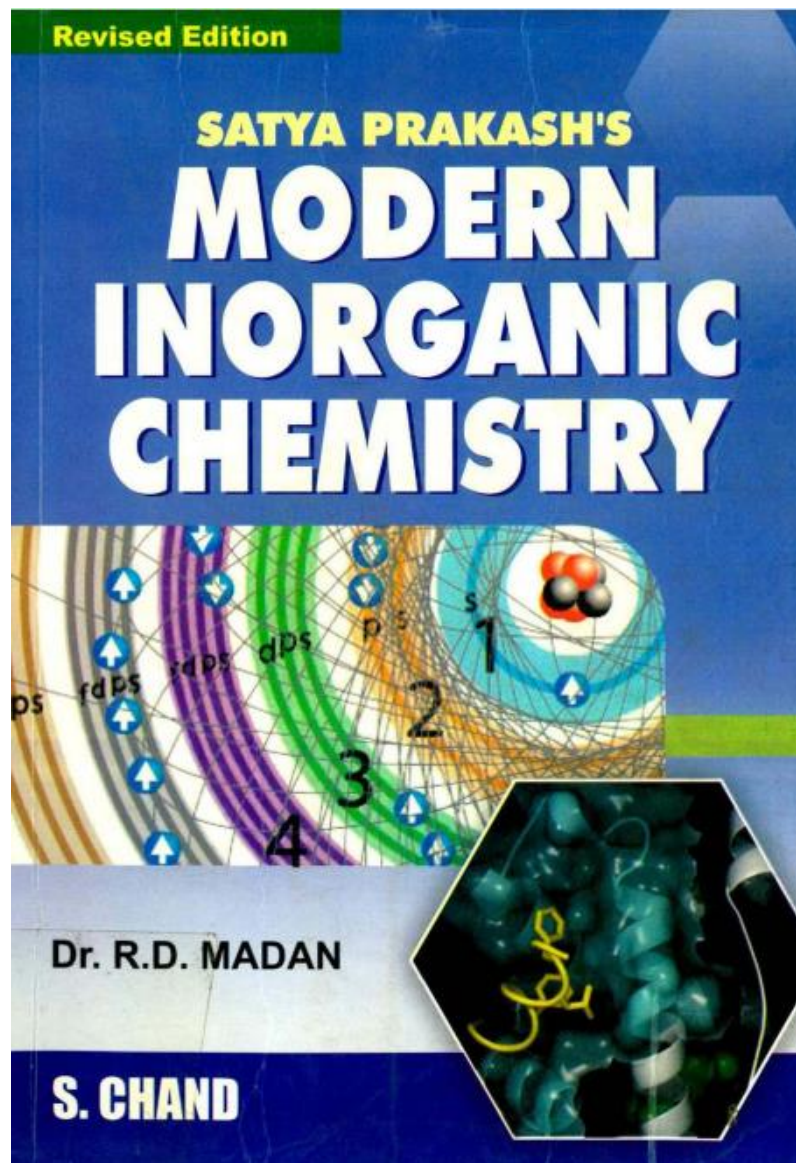


GENERAL CHEMISTRY – CHE 1201

Chemistry of Elements

d-block - Transition elements

Referencing Texts



The Transition Elements and Their Coordination Compounds

	1A (1)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
--	-----------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

The Transition Elements and Their Coordination Compounds

What are d-Block Elements?

Either in their atomic state or in any of their common oxidation state, the last electron enters the d-orbital of the inner shell, i.e., the last electron goes to $(n-1)d$ orbital. In other words, in these elements $(n-1)d$ orbitals are being progressively filled. Hence the name d-block elements.

Classification of d-Block Elements.

Based on whether the differentiating electron goes to 3d, 4d, 5d or 6d orbitals, d-block elements have been classified into four series, 3d, 4d, 5d and 6d.

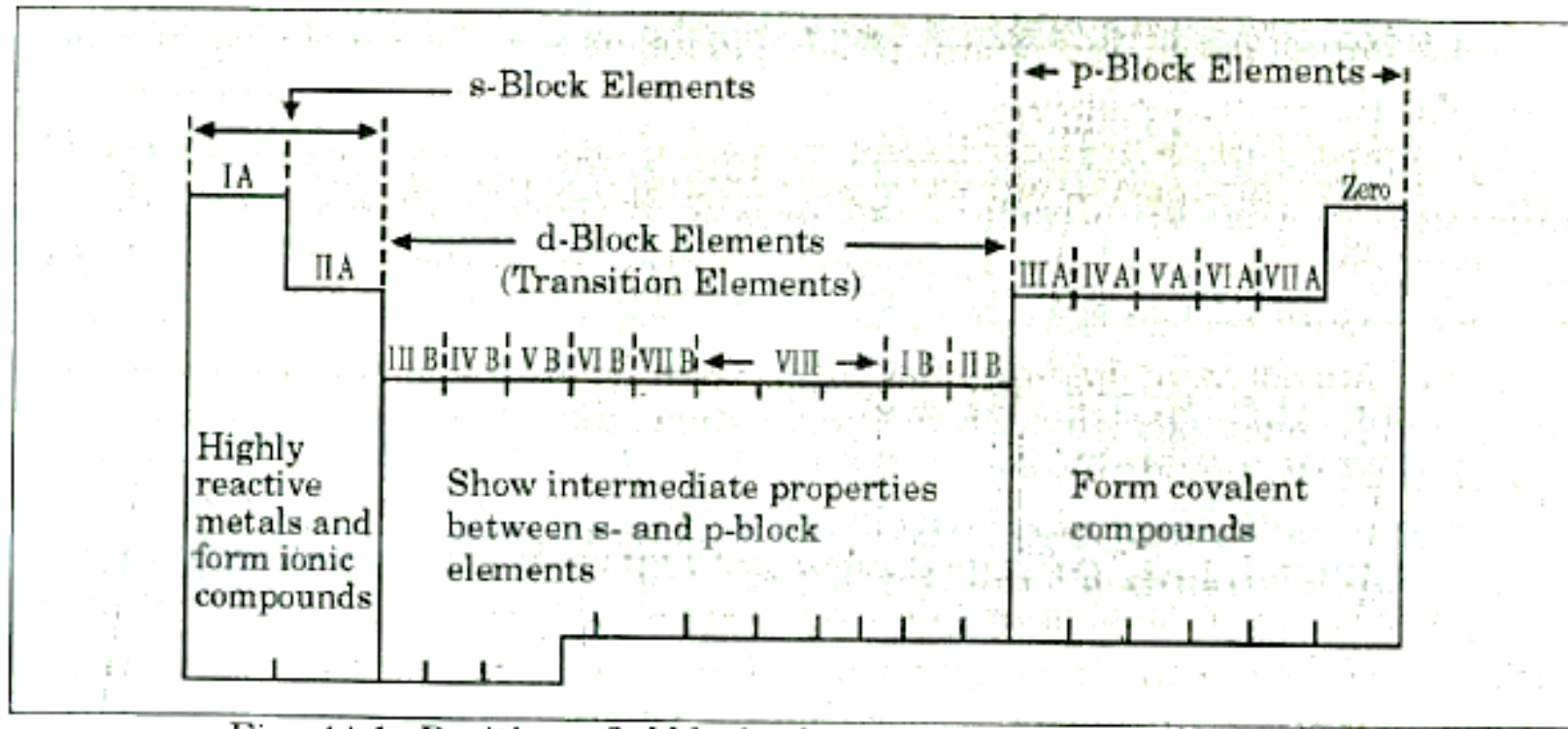


Fig. 44.1. Position of *d*-block elements in the periodic table.

Why are d-Block Elements Called Transition Elements?

The d-block elements are called transition elements because they exhibit transitional behaviour between highly reactive ionic-compound forming s-block elements (electropositive elements) on one side and mainly covalent-compound forming p-block elements (electronegative elements) on the other side.

General Characteristics of d-Block Elements

1. Metallic Character.

- All the transition elements are metals, since the number of electrons in the outermost shell is very small, being equal to 2.
- They are hard, malleable and ductile.
- Metals of VIII and IB groups are softer and more ductile than other metals.
- It appears that covalent and metallic bonding both exist in the atoms of transition metals. The presence of unfilled d-orbitals favours covalent bonding.
- These metals are good conductor of heat and electricity.

2. Atomic Volumes and Densities.

- The atomic volumes and densities of the elements of 3d-series are given below.

<i>Elements :</i>	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
<i>Atomic volume :</i> <i>(cm³)</i>	15.0	1.66	8.3	7.2	7.3	7.1	6.7	6.6	7.1	9.2
<i>Density (g/cm³) :</i>	3.0	4.5	6.1	7.2	7.4	7.8	8.9	8.9	8.9	7.1

- In d-block elements (n-1)d sub-shell is filled and the increased nuclear charge pulls the electrons in.
- This leads to the comparatively low atomic volumes of these elements compared to those of the neighbouring s- and p-block elements.
- The densities of d-block elements are very high in comparison to those of group IA and IIA elements. This is understandable because of their decreasing volumes.

3. Melting and Boiling Points.

- The melting and boiling points of the elements of 1st transition series are given below

<i>Elements :</i>	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
<i>Boiling point (°C) :</i>	2730	3260	3450	2665	2150	3000	2900	2730	2595	906
<i>Melting point (°C) :</i>	1539	1668	1900	1876	1245	1536	1495	1453	1083	419.5

- d-block elements have high melting and boiling points because they are held by strong forces.
- It can be seen from the table that the melting and boiling points have highest values in the middle of the series.
- In this region the elements have maximum number of unpaired d-electrons available for bonding.
- The elements also exhibit the highest oxidation states in this region.
- The trend in melting and boiling points shown by 3d-series elements is also followed by 4d and 5d-series elements.

<i>Elements :</i>	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
<i>Boiling point (°C) :</i>	2730	3260	3450	2665	2150	3000	2900	2730	2595	906
<i>Melting point (°C) :</i>	1539	1668	1900	1876	1245	1536	1495	1453	1083	419.5

- However, Zn, Cd and Hg have relatively low values

- melting and boiling points for these metals are

Zn = 419.5°C, 906°C

Cd = 320.9°C, 765°C

Hg = -38.4°C, 357°C

- The reason of these low values is that these metals have completely filled d-orbitals with no unpaired electron

Zn = $3d^{10}4s^2$ with $n = 0$

Cd = $4d^{10}4s^2$ with $n = 0$

Hg = $5d^{10} 6s^2$ with $n = 0$

- Those electrons may be available for covalent bonding amongst the atoms of these metals and is responsible for high melting and boiling points.

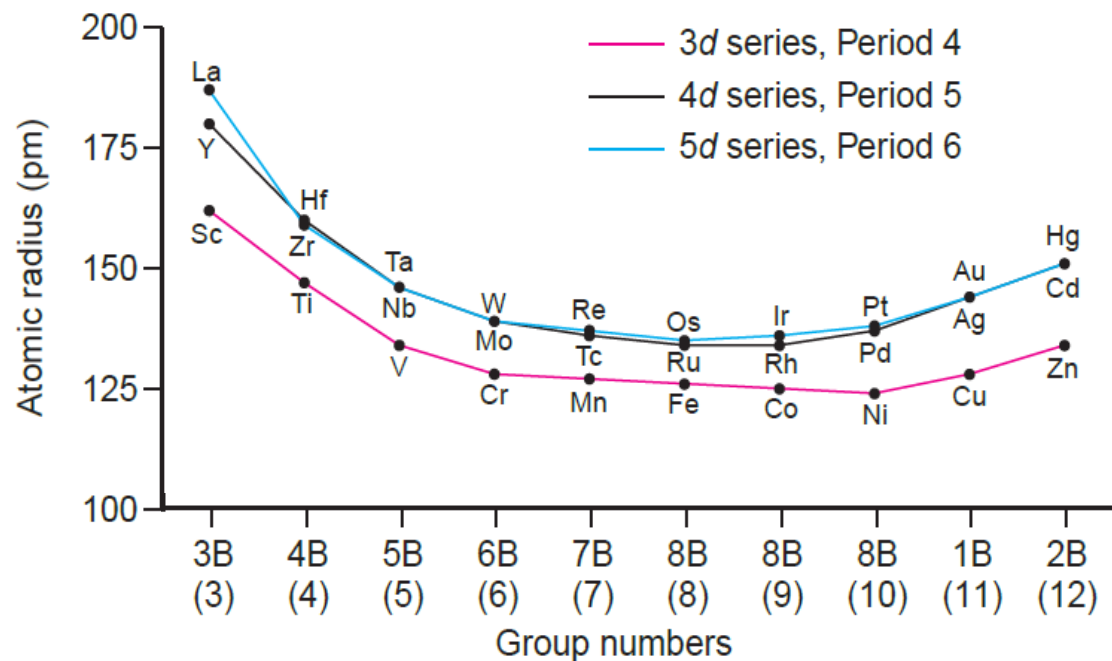
- Generally the melting and boiling points of d-block elements increase on descending a group. However, melting and boiling points of Ag are lower than those of Cu and Au.
- This can be explained on the basis that both Cu and Au have greater tendency to acquire higher oxidation state than +1, suggesting that in these metals the involvement of d-orbitals for metallic bonding is to a greater extent than in Ag.

4. Atomic (Covalent) Radii.

The atomic (covalent) radii of d-block elements are given below.

Sc 1.44 Å	Ti 1.32	V 1.22	Cr 1.18	Mn 1.17	Fe 1.17	Co 1.16	Ni 1.15	Cu 1.17	Zn 1.25
Y 1.62 Å	Zr 1.45	Nb 1.34	Mo 1.30	Tc 1.27	Ru 1.25	Rh 1.25	Pd 1.28	Ag 1.34	Cd 1.48
La 1.68 Å	Hf 1.44	Ta 1.34	W 1.30	Re 1.28	Os 1.26	Ir 1.27	Pt 1.30	Au 1.34	Hg 1.49

- These values generally decrease on moving from left to right in the series (i.e. period).
- This is due to the fact that as the nuclear charge increases with increasing atomic number across any series, the attraction of the electrons towards the nucleus increases and this leads to a decrease in the radii values across each period.



There are, however, a few exceptions.

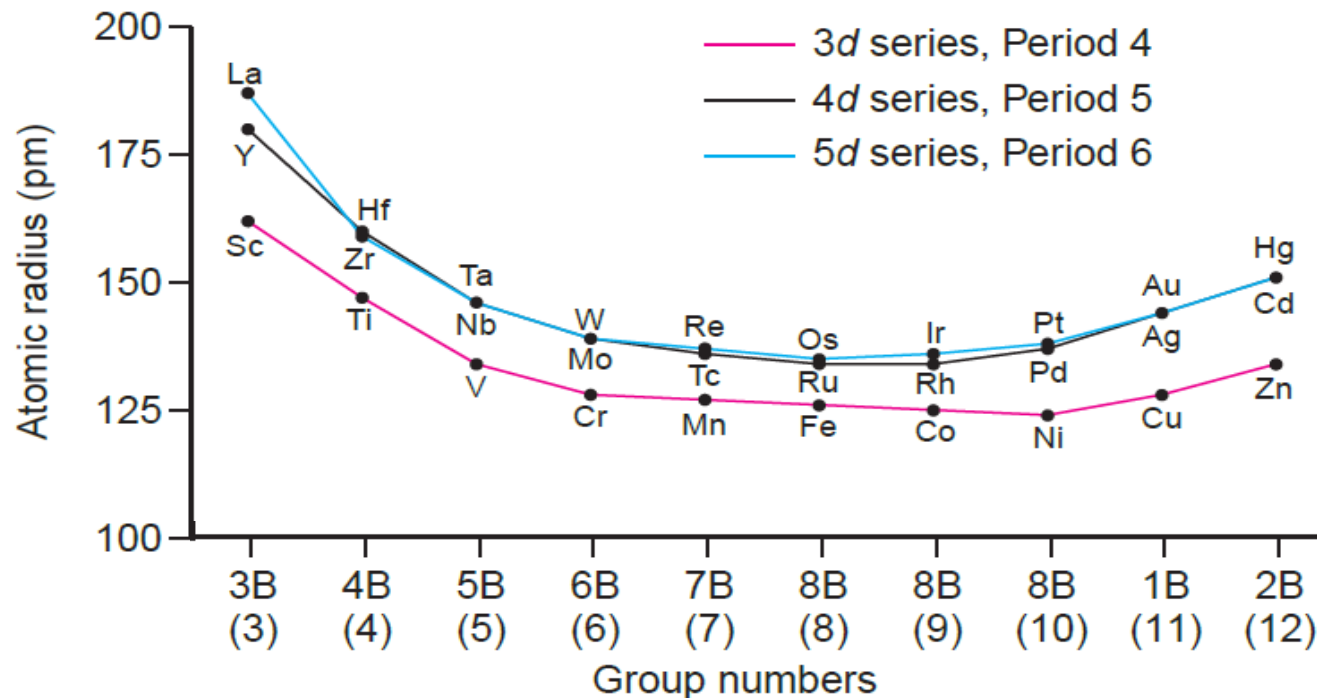
1. The atomic radii of the elements from Cr to Cu are very close to one another.

- As we move across the period, there is an increase in nuclear charge which tends to pull the 4s-electron inward near the nucleus, i.e. *it tends to reduce the size*.
- At the same time the addition of extra electrons to the 3d-orbitals also provides a screening effect which shields the 4s-electrons from the inward pull of the positive nucleus on these electrons. This screening effect increases as the number of extra electrons being added to 3d-orbitals increases. This effect *opposes the increase in size*.
- Thus there are operating two effects namely **screening effect** and the **nuclear charge effect** which oppose each other, resulting in that the effect of increase in the screening effect almost compensates the corresponding effect in increase in nuclear charge and hence very slight variations in the atomic radii from Cr to Cu are observed.

There are, however, a few exceptions.

2. In a particular series the atomic radius attains a minimum value for the elements of VIII group and increases again towards the end of the series.

- This is explained in terms of the increased force of repulsion among the added electrons which exceeds the attractive force due to the increased nuclear charge and results in the expansion of the electron cloud.



3. On descending a group n increases. The atomic size is, therefore, expected to increase from top to bottom.

- But this increase is not the same for all the members. The difference in radius of the second and third series elements is very small ($\sim 0.02 \text{ \AA}$) as compared to the first and second series members. This is due to **lanthanide contraction**.
- Due to the inclusion of 14 lanthanides between La ($Z = 57$, Group IIIB) and Hf ($Z = 72$, Group IV B), Hf is not as large as it would have been, if it had followed directly after La. The nuclear charge in the second series members are 18 units greater than the first series members, but this difference between the 2nd and 3rd series elements is 32. Thus, the increase in n [from 2nd series ($n = 5$) to 3rd series ($n = 6$)], though tends to increase the size, the excessive increase in nuclear charge annuls this effect. Consequently the elements of 2nd and 3rd series are very similar in size.

6. Ionization potentials.

- The ionization potential values of most of the d-block elements lie in between those of s- and p-block elements.
- This indicates that the d-block elements are less electropositive than s-block elements and more so than p-block elements.
- Thus d block elements do not form ionic compounds so readily as the alkali and alkaline earth metals do.
- Unlike s-block elements, d-block elements form covalent compounds as well.

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
1st	631	658	650	652.8	717.4	759.4	758	736.7	745.5	906.4
2nd	1235	1310	1414	1635	1509.1	1561	1646	1753	1957.9	1733.3
3rd	2389	2852	2828	2987	2248.4	2957.4	3232	3393	3554	3832.7

6. Ionization potentials.

From these values the following points may be noted

- The ionization potentials of 3d series elements increase as we move across the series from left to right, although the increase is not quite regular, e.g., the values for Sc, Ti, V and Cr differ very slightly.
- Similarly the values for Fe, Co, Ni and Cu are fairly close to one another.
- The value for Zn is appreciably higher due to the additional stability associated with completely-filled 3d-level in Zn
(Zn $\longrightarrow$ $3d^{10} 4s^2$).
- Slight variations that occur in ionization potentials across the series are mainly due to the slight changes in atomic radii which are on account of the fact that the screening effect caused by the addition of extra electrons to 3d-level almost compensates the effect in increase in nuclear charge, as we move from left to right in the series.

- As can be seen, second ionization energies for Cr and Cu are sufficiently higher than those of the neighbouring elements on the left as well as on the right.
- This is because the outer electronic configuration of Cr is $3d^5$ (half-filled) and that of Cu is $3d^{10}$ (completely-filled).
- Thus, the second ionization of chromium means the removal of one electron from the highly stable exactly half-filled sub-shell.
- Similarly, the second ionization of copper involves the removal of one electron from the highly stable completely-filled 3d sub-shell. This explains why the second ionization energies of Cr and Cu are higher than those of the neighbouring elements.

7. Different Oxidation States: Variable Valency

The term oxidation state is preferred to valency. Most of the transition metals show several oxidation states or variable valency in their compounds. Oxidation states shown by the elements of 3d-series are : Sc ($Z = 21$) $\rightarrow (+2), +3$; Ti ($Z = 22$) $\rightarrow +2, +3, +4$; V ($Z = 23$) $\rightarrow +2, +3, +4, +5$; Cr ($Z = 24$) $\rightarrow +2, +3, (+4), +6$; Mn ($Z = 25$) $\rightarrow +2, +3, +4, (+6), +7$; Fe ($Z = 26$) $\rightarrow +2, +3, (+4), (+6)$; Co ($Z = 27$) $\rightarrow +2, +3, (+4)$; Ni ($Z = 28$) $\rightarrow +2, (+3)$; Cu ($Z = 29$) $\rightarrow +1, +2$; Zn ($Z = 30$) $\rightarrow +2$. Oxidation states shown in brackets are less common oxidation states, *i.e.* these oxidation states are exhibited in a few compounds.

7. Different Oxidation States: Variable Valency

- The cause of showing different oxidation states is that these elements have several $(n-1)d$ and ns electrons and that the energies of $(n-1)d$ and ns orbitals are fairly close to each other. Hence $(n-1)d$ electrons are as easily lost as ns electrons. The following points may be noted;
 - i. Among the elements of 3d series, Mn shows the maximum number of oxidation states because it is at Mn that the number of unpaired electrons in 3d orbitals is maximum, On proceeding towards right or left from Mn, the number of unpaired electrons in 3d orbitals decreases and hence the number of oxidation states shown by the elements lying at the right or left of Mn decreases.
 - ii. With the exception of Cr ($3d^5 4s^1$), the minimum oxidation state shown by the elements of 3d-series is equal to the number of electrons in 4s orbital.
 - iii. Excepting the elements of group IB, other elements show maximum oxidation state which is never more than their group number.

iv. In going down a group, the higher oxidation states become more stable. For example +2, and +3 oxidation states shown by Fe are stable oxidation states while +4 and +6 are unstable ones. Similarly +2, +3, +4, and +6 are stable while +5, +7 and +8 are unstable oxidation states shown by Ru.

v. The relative stability of various oxidation states of a given element can be explained on the basis of the stability of d^0 , d^5 and d^{10} configurations. For example

- a. Ti^{4+} ($3d^{10} 4s^0$) is more stable than Ti^{3+} ($3d^1 4s^0$), because of the presence of $3d^{10}$ orbital in Ti^{4+} , ion.
- b. Mn^{2+} ($3d^5 4s^0$) is more stable than Mn^{3+} ($3d^4 4s^0$), since Mn^{2+} ion has $3d^5$ orbital.
- c. Ag^+ ($4d^{10} 5s^0$) is more stable than Ag^{2+} ($4d^9 \& 5s^0$) because of the presence of $4d^{10}$ orbital in Ag^+ ion.
- d. Transition elements can form ionic bonds in their lower oxidation states and the ionic character of the bond decreases (or covalent character of the bond increases) with the increase of oxidation state. With the decrease of ionic character, the acidic character of oxides or chlorides increases.

(a) With the decrease of the ionic character of M—O bond in the oxides, the acidic character of the oxides increases as shown below :

←——— Oxidation states ———→			
+2	+3	+4	+6
TiO	Ti ₂ O ₃	TiO ₂	—
VO	V ₂ O ₃	—	—
CrO	Cr ₂ O ₃	—	CrO ₃
MnO	MnO ₃	MnO ₂	MnO ₃ (unstable)
Basic	Amphoteric		Acidic
— Oxidation state increasing —→			
— Ionic character of M—O bond decreasing —→			
— Acidic character of oxides increasing —→			

(b) With the decrease of the ionic character of M—Cl bond in the chlorides, the acidic character of the chlorides increases as shown below :

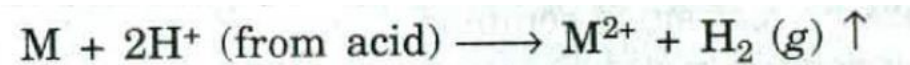
←——— Oxidation states ———→			
+2	+3	+4	+6
TiCl ₂	TiCl ₃	TiCl ₄	—
VCl ₂	VCl ₃	VCl ₄	VOCl ₃
— Oxidation state increasing —→			
— Ionic character of M—Cl bond decreasing —→			
— Acidic character of chlorides increasing —→			

8. Standard Reduction Potentials and Reducing Properties.

Standard reduction potentials (in volts) of some of the d-block elements are given below. These values are given in the decreasing order.

E° (Volts)		E° (Volts)	
$\text{Hg}_2^{2+} + 2e^- \longrightarrow 2\text{Hg} ;$	+ 0.79	$\text{Cr}^{3+} + 3e^- \longrightarrow \text{Cr} ;$	- 0.71
$\text{Cu}^{2+} + 2e^- \longrightarrow \text{Cu} ;$	+ 0.34	$\text{Zn}^{2+} + 2e^- \longrightarrow \text{Zn} ;$	- 0.76
$\text{Ni}^{2+} + 2e^- \longrightarrow \text{Ni} ;$	- 0.24	$\text{Mn}^{2+} + 2e^- \longrightarrow \text{Mn} ;$	- 1.05
$\text{Co}^{2+} + 2e^- \longrightarrow \text{Co} ;$	- 0.28	$\text{V}^{2+} + 2e^- \longrightarrow \text{V} ;$	- 1.20
$\text{Cd}^{2+} + 2e^- \longrightarrow \text{Cd} ;$	- 0.40	$\text{Ti}^{2+} + 2e^- \longrightarrow \text{Ti} ;$	- 1.60
$\text{Fe}^{2+} + 2e^- \longrightarrow \text{Fe} ;$	- 0.44	$\text{Sc}^{3+} + 3e^- \longrightarrow \text{Sc} ;$	- 2.08
		$\text{La}^{3+} + 3e^- \longrightarrow \text{La} ;$	- 2.37

- These values make the following points evident;
 - (i) Excepting for Hg and Cu, these values for other elements are negative, i.e. less than that of hydrogen whose value is taken as zero. Their values being less than that of hydrogen, make these metals able to liberate hydrogen from dilute acids.



- However, most of these metals react with acid and liberate H_2 gas is very slow.
- Some of the metals get protected from further attack of the acids due to the formation of a thin protective layer of their oxide which prevents the acid to come in further contact with the metal,

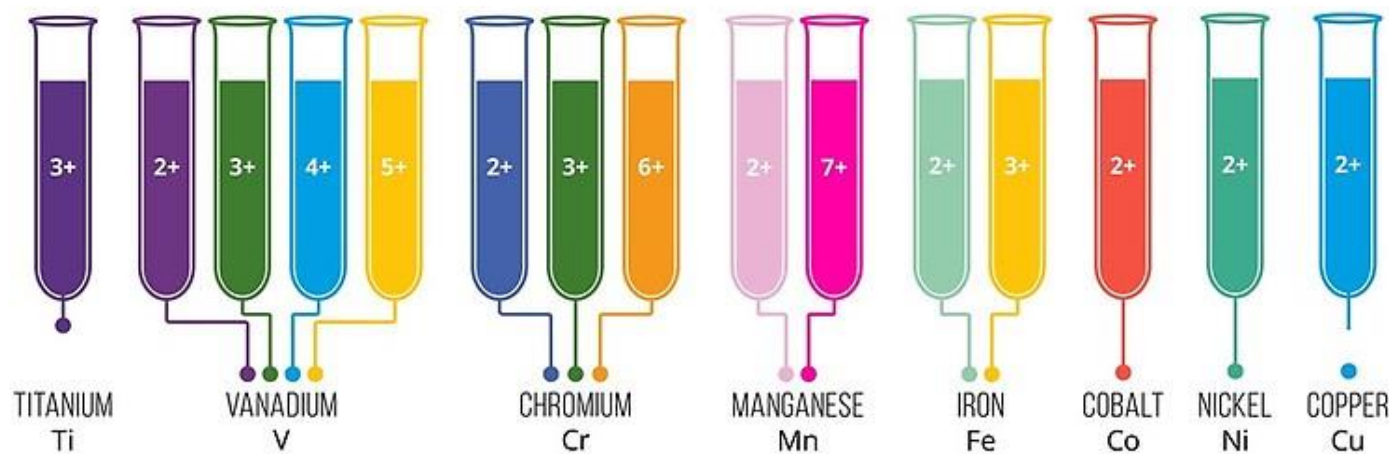
e.g., although the electrode potential of Cr is much less than that of hydrogen ($E^0 Cr^{3+}/Cr = - 0.71$ volt), it gets coated with its non-reactive oxide, Cr_2O_3 which makes it so unreactive that can be used as a protective non-oxidizing metal.

- ii.
 - In spite of the fact that all the metals of first transition series, excepting Cu, have high negative values of standard electrode potentials, these are not as good reducing agents as the metal of groups IA, IIA and IIIA.
 - The poor reducing capacity of the transition metals is due to high heats of vaporization, high ionization potentials and low heats of hydration of their ions.

9. Formation of Coloured Compounds.

Normally the compounds of *s*- and *p*-block elements are colourless while those of transition metals in the solid or in solution state are usually coloured. The colours of hydrated cations of transition metals of the first transition series are : Sc^{3+} , Ti^{4+} ($3d^0$, $n = 0$) $\rightarrow$ colourless; Ti^{3+} ($3d^1$, $n = 1$) $\rightarrow$ purple; V^{3+} ($3d^2$, $n = 2$) $\rightarrow$ green; V^{2+} ($3d^3$, $n = 3$) $\rightarrow$ violet; Cr^{3+} ($3d^3$, $n = 3$) $\rightarrow$ green; Cr^{2+} ($3d^4$, $n = 4$) $\rightarrow$ blue ; Mn^{3+} ($3d^4$, $n = 4$) $\rightarrow$ violet; Mn^{2+} ($3d^5$, $n = 5$) $\rightarrow$ faintly pink or faintly green; Fe^{3+} ($3d^5$, $n = 5$) $\rightarrow$ yellow; Fe^{2+} ($3d^6$, $n = 4$) $\rightarrow$ pale green or faintly green; Co^{3+} ($3d^6$, $n = 4$) $\rightarrow$ blue; Co^{2+} ($3d^7$, $n = 3$) $\rightarrow$ pink; Ni^{2+} ($3d^8$, $n = 2$) $\rightarrow$ green; Cu^{2+} ($3d^9$, $n = 1$) $\rightarrow$ blue; Cu^+ ($3d^{10}$, $n = 0$) $\rightarrow$ colourless; Zn^{2+} ($3d^{10}$, $n = 0$) $\rightarrow$ colourless.

THE COLOURS OF AQUEOUS TRANSITION METAL IONS



Here n denotes the number of unpaired electrons in the respective cations. For simplicity water molecules associated with each cation have not been shown.

- It may be seen from the colours given above that the cations (e.g. Sc^{3+} , Ti^{4+} , Cu^+ , Zn^{2+}) having vacant or completely-filled d-orbitals (d^0 and d^{10} orbitals) are colourless while those (e.g. Ti^{3+} , V^{3+} , Cr^{3+} etc.) having partially-filled d-orbitals (d^1 , d^2 , d^3 etc. orbitals) are coloured.
- In other words, the cations having all the electrons paired in d-orbitals (i.e. completely-filled d-orbitals) or no electrons in d-orbitals (i.e. empty d-orbitals) are colourless while those having some or all the electrons unpaired in d-orbitals are coloured.
- If n is the number of unpaired electrons in d-orbitals, then the ions having $n = 0$ are colourless while those having $n = 1, 2$ etc. are coloured.

10. Magnetic Properties.

There are, in general, three types of magnetism:

- (i) Paramagnetism
- (ii) Diamagnetism
- (iii) Ferromagnetism

Paramagnetism.

When a substance is placed between the magnetic poles, it allows the magnetic lines of force to pass through them rather than through vacuum, i.e. a paramagnetic bar will tend to set itself with its length parallel to the field, when it is placed between the magnetic poles. In other words a paramagnetic substance is attracted into a magnetic field.

- Paramagnetism is common among the transition metal complexes, i.e. most of the transition metals are paramagnetic.

6. Chemical Properties of the Transition Metals

Oxidation States

- One of the most characteristic chemical properties of the transition metals is the occurrence of multiple oxidation states. For example, in their compounds, vanadium exhibits two common positive oxidation states, chromium three, and manganese three.
- Since the ns and $(n - 1)d$ electrons are so close in energy, transition elements can use all or most of these electrons in bonding. This behavior is markedly different from that of the main-group metals, which display one or, at most, two oxidation states in their compounds.



A

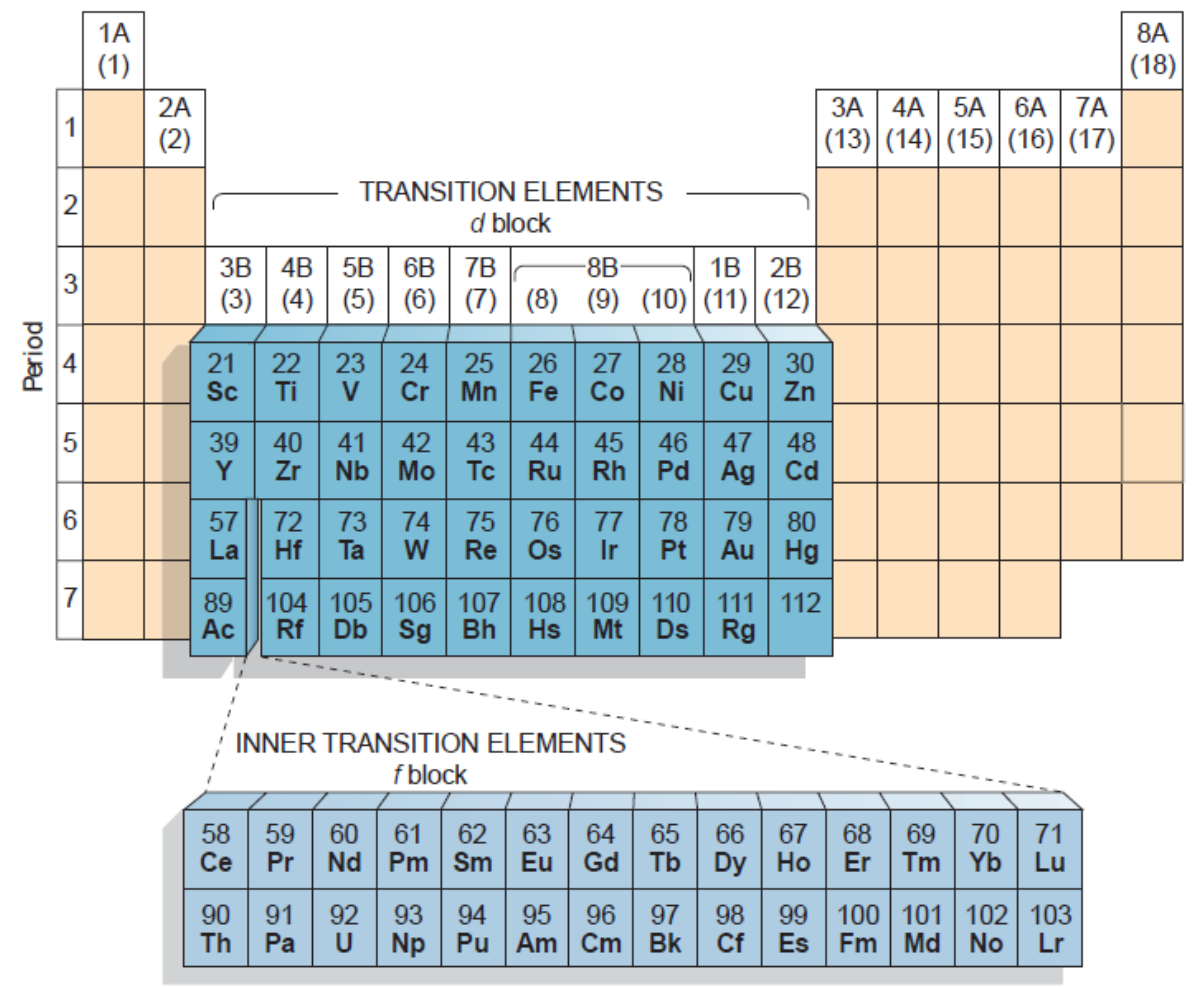
Figure 23.5 Aqueous oxoanions of **transition** elements. **A**, Often, a given **transition** element has multiple oxidation states. Here, Mn is shown in the +2 (Mn^{2+} , *left*), the +6 (MnO_4^{2-} , *middle*), and the +7 states (MnO_4^- , *right*).



B

B, The highest possible oxidation state equals the group number in these oxoanions: +5 in VO_4^{3-} (*left*), +6 in $\text{Cr}_2\text{O}_7^{2-}$ (*middle*), and +7 in MnO_4^- (*right*).

The Transition Elements and Their Coordination Compounds



Scandium, Sc; 3B(3)



Titanium, Ti; 4B(4)



Vanadium, V; 5B(5)



Chromium, Cr; 6B(6)



Manganese, Mn; 7B(7)

Figure 23.1 The transition elements (*d* block) and inner transition elements (*f* block) in the periodic table.

Electron Configurations of the Transition Metals and Their Ions

Table 23.1 Orbital Occupancy of the Period 4 Transition Metals

Element	Partial Orbital Diagram										Unpaired Electrons
	4s	3d					4p				
Sc	↑↓	↑								1	
Ti	↑↓	↑	↑							2	
V	↑↓	↑	↑	↑						3	
Cr	↑	↑	↑	↑	↑	↑				6	
Mn	↑↓	↑	↑	↑	↑	↑				5	
Fe	↑↓	↑↓	↑	↑	↑	↑				4	
Co	↑↓	↑↓	↑↓	↑	↑	↑				3	
Ni	↑↓	↑↓	↑↓	↑↓	↑	↑				2	
Cu	↑	↑↓	↑↓	↑↓	↑↓	↑↓				1	
Zn	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓				0	